UPPSALA
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MASTER THESIS SPECIFICATION

Implicit Neural Representations for Speed-of-Sound Imaging

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1 Title

This thesis project explores learning based models for ultrasound image reconstruction based on implicit neural representation (INRs) titled with "Implicit Neural Representations for Speed-of-Sound Imaging".

2 Abstract

Medical ultrasound is a primary diagnostic modality, yet conventional B-mode imaging offers only qualitative contrast based on echo amplitudes. Speed-of-Sound (SoS) imaging has emerged as a quantitative alternative, providing biomarkers capable of distinguishing pathological tissue changes. However, reconstructing SoS maps from limited-aperture hand-held probes is an ill-posed inverse problem. Existing reconstruction methods rely on discrete grid-based representations and simplified straight-ray propagation models, which inherently suffer from resolution trade-offs and fail to account for non-linear wave physics such as refraction.

This thesis proposes a novel reconstruction framework based on Implicit Neural Representations (INRs). Unlike traditional grid-based methods, this approach models the tissue property map as a continuous function parameterized by a coordinate-based neural network. By leveraging Fourier feature embeddings to capture high-frequency spatial details and integrating a joint space optimization framework, the network is optimized in a self-supervised manner. The project aims to demonstrate that INRs can surpass the performance of linear baselines, offering resolution-independent reconstruction and superior handling of complex tissue heterogeneities for clinical applications such as breast cancer imaging.

3 Background

Medical ultrasound is ubiquitous in clinical practice due to its non-ionizing nature, cost-effectiveness, real-time, and portability. However, conventional B-mode imaging relies exclusively on the amplitude of backscattered echoes (the signal envelope), effectively discarding the rich phase and transmission information encoded in the raw ultrasound signal (Oelze and Mamou, 2016). Consequently, B-mode remains a qualitative modality, its output depends on system settings (such as gain) rather than intrinsic tissue properties, leading to diagnostic shortcomings in various clinical applications (Bezek *et al.*, 2025).

To overcome these limitations, quantitative approaches such as Speed-of-Sound (SoS) imaging have emerged. By reconstructing maps of local acoustic wave velocity (Rau *et al.*, 2021), SoS imaging exploits the fundamental link between tissue biomechanics and wave propagation. Since pathological processes, such as malignant tumor formation or hepatic steatosis, alter the bulk modulus and density of tissue, they induce measurable deviations in

sound speed. Therefore, SoS maps serve as a direct, quantitative biomarker, offering superior tissue characterization compared to standard amplitude-based contrast.

To enable SoS imaging using standard hand-held clinical transducers (where double-sided access to tissue is unavailable), reconstruction methods rely on disparity estimation from a pair of transmissions. This involves measuring time-shifts between echoes observed from different insonification directions to infer the underlying sound speed distribution. Mathematically, this constitutes a linear inverse problem where the observed disparities are linked to the SoS map via a forward imaging model (system matrix). Traditionally, this imaging model is constructed using the straight-line propagation assumption (Ray Theory). However, this assumption simplifies the underlying physics, neglecting wave broadening (diffraction) and the bending of sound rays (refraction), leading to reconstruction artifacts. While recent studies (Bezek *et al.*, 2025) have proposed learning effective wave propagation models via convolutional operators, these methods often remain bound to fixed pixel grids and specific plane-wave geometries.

4 Description of Tasks

The central problem addressed in this thesis is the reconstruction of quantitative Speed-of-Sound (SoS) maps from limited-aperture ultrasound data, a classic ill-posed inverse problem, using hand-held transducers. Current state-of-the-art methods, such as the disparity-based linear corrections proposed by Bezek *et al.*, (2025), rely on discrete grid-based representations. These methods are inherently limited by the trade-off between spatial resolution and computational memory, and they often struggle to model the continuous nature of wave physics, leading to discretization artifacts around complex tissue boundaries.

The core task of this project is to formulate and solve this inverse problem using Implicit Neural Representations (INRs**)**. Unlike traditional approaches that optimize a pixel grid, this project will analyze how a coordinate-based neural network (parameterized by weights θ) can learn the continuous function mapping spatial coordinates (x, y) to physical tissue properties (sound speed). The interesting analytical challenge lies in replacing the explicit system matrix inversion with a neural optimization loop, utilizing continuous integration to maximize resolution

The project involves the following distinct phases of work:

Data Generation and Linear Baseline. Utilize acoustic simulation tools (e.g., k-Wave Matlab/C++ simulating framework) to generate a dataset of ground-truth SoS maps and corresponding ultrasound disparity (time-delay) measurements. The method will employ virtual source (VS) transmission sequences, as circular wavefronts provide superior geometric constraints for high-resolution INR reconstruction (Wang *et al.*, 2026) and higher SoS reconstruction accuracy compared to plane waves (Rau *et al.*, 2021). Consequently, since the convolutional formulation in (Bezek *et al.*, 2025) is specialized for plane wave physics, the project will implement conventional regularized least squares as the primary benchmark. This

establishes the standard grid-based performance ceiling, which the INR method aims to surpass by resolving sub-grid details through continuous representation..

INR Architecture and Joint Space Formulation. Design and implement the INR architecture. To mitigate the error propagation typical of sequential reconstruction methods (where noisy displacement estimates degrade the final image), we will implement a Joint Space Optimization framework consisting of two parallel INRs:

1. $INRsos(x, y)$**.** Parameterizes the quantitative Speed-of-Sound map (Anatomy).
2. $INRdisp(u, v)$: Parameterizes the observed Time-Delay/Displacement field (Data). This involves selecting appropriate positional embeddings (e.g., Fourier features) to enable the networks to capture high-frequency details.

Optimization and Continuous Physics Integration. Develop the optimization loop to train the INRs.The loop will minimize a dual objective: ensuring $INRdisp$ approximates the measured ultrasound data (denoising) while simultaneously satisfying the physics constraints imposed by projecting $INRsos$ through the forward model. Instead of using a fixed grid-based matrix L, we will implement a Differentiable Ray Integration scheme. This involves sampling the $INRsos$ at continuous coordinates along the acoustic ray paths and integrating the slowness to predict time delays, allowing for "grid-free" reconstruction limited only by acoustic wavelength.

Evaluation and Clinical Application. Quantitatively evaluate the INR-based reconstruction against the grid-based regularized least squares baseline using metrics such as Root Mean Square Error (RMSE) and Contrast-to-Noise Ratio (CNR) on simulation data. Apply the developed INR framework to real clinical data from hand-held breast ultrasound scans. Assess if the continuous nature of INRs provides superior visualization of lesion boundaries compared to discrete grid reconstructions.

5 Methods

Systems and Tools. The project will primarily use Python as the development language. The deep learning framework PyTorch will be used to implement the Implicit Neural Representations and the automatic differentiation engines required for the optimization loop. For acoustic simulation and data generation, MATLAB and the k-Wave toolbox will be utilized.

Methodological Approach. The reconstruction methodology shifts from standard supervised learning (training on a large database) to test-time optimization. The method utilizes fully connected neural networks to represent the image volume and displacement fields. To ensure the network can capture fine details rather than over-smoothed approximations, the input coordinates will be transformed using Fourier feature embeddings before passing through the MLP, as demonstrated by Tancik *et al.*, (2020) to improve the convergence in coordinate based tasks. The optimization will be driven by a physics-informed loss function that

penalizes the discrepancy between the smoothed displacement field ($INRdisp$) and the physics-predicted delays derived from continuously integrating the anatomy ($INRsos$). This Continuous Differentiable Integration allows the model to resolve features at sub-pixel scales that would otherwise be lost in grid-based forward models.

Evaluation and Documentation. Results will be evaluated quantitatively using synthetic data where the ground truth SoS maps are known. Key performance metrics will include Root Mean Square Error (RMSE) to measure global accuracy and Contrast-to-Noise Ratio (CNR) to assess the method's ability to distinguish lesions from background tissue. Qualitatively, the method will be evaluated on real clinical breast ultrasound data. Specifically, we will assess the "detectability" of lesions for which standard B-mode images are inconclusive but exhibit distinct stiffness properties. Success will be defined by the INR's ability to provide diagnostic information for these structures where B-mode fails. The work will be documented in a comprehensive thesis report and the source code will be version-controlled and delivered via a repository.

6 Relevant Courses

- Computer Programming II - 1TD722
- High Performance Programming - 1TD062
- Linear Algebra for Data Science - 1MA330
- Artificial Intelligence for Industrial Analytics - 1TS321
- Statistical Machine Learning - 1RT700
- Advanced Probabilistic Machine Learning - 1RT705
- Optimisation - 1TD184
- Data Engineering I - 1TD169
- Data Engineering II - 1TD076

7 Delimitations

This thesis focuses exclusively on Learning-based Models for Ultrasound Image Reconstruction using Implicit Neural Representations (INRs). It is limited to 2D imaging scenarios using data derived from simulation and hand-held linear probes;

While 3D volumetric reconstruction and real-time hardware deployment are outside the scope of this work, the project will benchmark the computational reconstruction speed (latency per frame) on a standard GPU workstation. This ensures the proposed method is evaluated for clinical feasibility alongside image quality.

8 Time Plan

This thesis work is expected to be done in 20 weeks, beginning in the fourth week of January 2026. The completion is aimed to be in the 21st week of 2026, upon a public presentation regarding the result as well as revision of the report based on the comments by the audience and reviewer. I have other commitments such as teaching and research assistance at the IT department of Uppsala University and consultancy responsibility on weekends. I had these commitments when I started the master's program and they haven't affected my studies for the last one and half years and I don't expect them to do so during the thesis project. The preliminary time allocation is detailed in the following table and attached Gantt chart.

Phase	Duration (Weeks)	Time Plan	Key Milestones and Deliverables	Proposed meeting schedule with Orcun and Deniz
Phase 1: Setup and Baseline	4 Weeks	W4 - W7	**[M1]: Literature review, setting compute environment, establishing linear baseline.**	
1.1 Focused Literature Review	1	W4	Focused review on INRs (SIREN, Fourier Features) and their application in inverse problems.	Meeting1 : Kick-off Haben Deniz Orcun
1.2 Environment Setup & Data Generation	1	W5	Configure PyTorch, k-Wave. Generate initial synthetic dataset.	
1.3 Implement Linear Baseline	2	W6 - W7	Implement the Regularized Least Squares reconstruction to establish the benchmark. Start writing the report.	M2: Progress report 1 Haben Deniz Orcun
Phase 2: INR Architecture & Drafting	5 Weeks	W8 - W12	**[M2]: Core architecture designed and theoretical chapters drafted.**	
2.1 INR Architecture Design & Forward Model	3	W8 - W10	Implement parallel MLPs (SoS and Displacement). Formulate the continuous integration layer.	M3: Progress report 2 Haben Deniz
2.2 Forward Model Formulation	2	W11-W12	Develop the differentiable link between the continuous INR output and the discrete disparity measurements. Define the initial loss function.	M4: Progress report 3 Haben Deniz Orcun
Phase 3: Optimization and Simulation Evaluation	7 Weeks	W13 - W19	**[M3]: Achieve robust, high-quality reconstruction on simulated data.**	

3.1 Initial Optimization & Tuning	3	W13 - W15	Implement the self-supervised, per-patient training loop. Address initial convergence and stability issues.	M5: Progress report 4 Haben Deniz
3.2 Hyperparameter Tuning & Analysis	2	W16 - W17	Compare different Positional Encodings and analyze the implicit regularization properties of the INRs.	M6: Progress report 5 Haben Deniz Orcun
3.3 Rigorous Simulation Evaluation	2	W18 - W19	Final quantitative comparison of the INRs method vs. the Linear Baseline (RMSE, CNR) across various phantom types.	M7: Progress report 6 Haben Deniz Orcun
Phase 4: Clinical Translation and Final Report	2 Weeks	W20 - W21	**[M4]: Demonstrate utility on real data and finalize the written thesis.**	
4.1 Clinical Data Application	1	W20	Pre-process and load real RF clinical data. Apply the best-performing INR model.	M8: Progress report 7 Haben Deniz Orcun
4.2 Final Thesis Compilation	1	W20 - W21	Write Results, Discussion, and Conclusion chapters. Integrate all written components and submit a near-final draft.	

References

Bezek, C.D. *et al.* (2025) “Learning the Imaging Model of Speed-of-Sound Reconstruction via a Convolutional Formulation,” *IEEE Transactions on Medical Imaging*, 44(9), pp. 3600–3609. Available at: https://doi.org/10.1109/TMI.2024.3480690.

Oelze, M.L. and Mamou, J. (2016) “Review of Quantitative Ultrasound: Envelope Statistics and Backscatter Coefficient Imaging and Contributions to Diagnostic Ultrasound,” *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, 63(2), pp. 336–351. Available at: https://doi.org/10.1109/TUFFC.2015.2513958.

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